

Predicting Crop Yield and Revenue for Data-Driven Agricultural Planning in Canada

Project Overview

Canadian agricultural producers must make planting decisions before knowing exactly how much a crop will yield or what price it will receive after harvest. These decisions are further complicated by differences in regional growing conditions, changing weather patterns, historical price volatility, and differences in crop productivity. Choosing a crop based only on its historical average yield or price may therefore lead to poor resource-allocation decisions.

The goal of this project is to develop a two-stage machine learning system that predicts **crop yield** and **average farm price** using historical Canadian agricultural and climate data. These predictions will then be combined to estimate **expected gross revenue per hectare** for different crops. Historical variability in yield, price, and revenue will also be analyzed to provide a measure of risk alongside expected return.

The intended client is a **Canadian agricultural producer** seeking to make more informed crop-planning and land-allocation decisions.

Business Problem

Agricultural producers must decide which crops to plant and how much land to allocate to them while facing uncertainty in both crop productivity and market prices. Crop yields can vary because of geographic and environmental conditions, while commodity prices fluctuate over time. A crop with a high expected yield may not necessarily generate the greatest revenue, and a crop with high expected revenue may carry substantially greater historical risk.

This project will address the following primary business question:

How can historical agricultural, climate, and pricing data be used to forecast crop yield and farm prices so that producers can compare expected gross revenue and risk when making planting decisions?

Rather than predicting total farm value directly, the project will separately model the two uncertain components that contribute to revenue: crop yield and crop price. This avoids creating a model that simply reconstructs farm value from variables such as realized production and price.

Stakeholders

The primary stakeholders are **Canadian agricultural producers and farm managers** responsible for crop selection and land allocation.

These stakeholders could use the model's results to:

- Compare the expected economic performance of different crops.
- Estimate expected yield under regional and climatic conditions.
- Estimate future farm prices using historical market information.
- Compare expected gross revenue per hectare across crops.
- Evaluate the tradeoff between expected revenue and historical volatility.
- Make more data-informed planting and resource-allocation decisions.

Secondary stakeholders could include agricultural consultants, farm-management organizations, lenders, and other organizations involved in agricultural planning.

Project Goals and Business Outcomes

The primary goal is to develop a decision-support framework that converts historical agricultural and climate data into actionable estimates of future crop performance.

The project will:

1. Develop a regression model to predict **crop yield in kilograms per hectare**.
2. Develop a second regression model to predict **average farm price in dollars per tonne**.
3. Combine predicted yield and predicted price to calculate **expected gross revenue per hectare**.
4. Measure historical variability in yield, price, and/or revenue to represent the relative risk associated with different crops.
5. Compare crops based on both expected revenue and historical stability.
6. Identify the variables that contribute most strongly to model predictions.

7. Translate model results into recommendations that can be understood by agricultural decision-makers.

The project will focus on **gross revenue rather than profit** because the available datasets do not contain sufficient information about production costs such as labor, fuel, fertilizer, seed, machinery, or transportation.

Why Machine Learning Is Appropriate

Crop yield and agricultural prices are influenced by multiple interacting factors and can change over time. Relationships between climate, location, crop type, historical performance, and agricultural outcomes may also be nonlinear.

Machine learning regression methods are appropriate because they can identify patterns across these variables and generate predictions for continuous outcomes such as yield and price.

Multiple algorithms will be compared rather than assuming that a single model is best. Simpler regression methods will provide interpretable baselines, while tree-based ensemble methods will allow nonlinear relationships and interactions to be captured.

Machine learning is particularly useful for the proposed decision system because predictions from the yield and price models can be combined to estimate the expected economic outcome of different planting scenarios.

Data Sources

Dataset 1: Farm Produce Data — 80 Years

Source: Kaggle — *Farm Produce Data | 80 Years*

The agricultural dataset contains **10,273 observations** and **9 variables** describing historical Canadian crop production by year, province, and crop.

The variables are:

- `REF_DATE` — Year
- `GEO` — Province
- `Type of crop` — Crop grown
- `Average farm price` (dollars per tonne)
- `Average yield` (kilograms per hectare)
- `Production` (metric tonnes)
- `Seeded area` (acres)
- `Seeded area` (hectares)
- `Total farm value` (dollars)

The records begin as early as **1908**, providing a long historical view of agricultural productivity and economic performance.

The dataset contains relatively little missing data in the main price, yield, and production variables. Seeded-area variables contain more missing observations and will require additional investigation during data preparation.

This dataset is directly relevant to the business problem because it contains the two primary prediction targets: **crop yield** and **average farm price**. It also contains the crop, geographic, production, land-use, and financial variables needed for exploratory and risk analysis.

Dataset 2: Eighty Years of Canadian Climate Data

Source: Kaggle — *Eighty Years of Canadian Climate Data*

The climate dataset contains **29,221 daily observations and 27 variables**.

It contains daily:

- Mean temperature
- Total precipitation

for 13 Canadian locations:

- Calgary
- Edmonton
- Halifax
- Moncton
- Montreal
- Ottawa
- Quebec City
- Saskatoon
- St. John's
- Toronto
- Vancouver
- Whitehorse
- Winnipeg

The climate records begin in **1940**. Coverage differs by location, particularly during earlier years, and several locations contain substantial missing values.

Rather than joining daily weather observations directly to annual agricultural records, the climate data will first be aggregated into meaningful **growing-season climate features**.

Potential engineered features include:

- Average growing-season temperature
- Total growing-season precipitation
- Temperature variability
- Number of unusually hot days
- Number of unusually cold days
- Number of precipitation or dry days

Weather stations will then be mapped to their corresponding province, allowing annual climate features to be joined to agricultural observations using **province and year** where appropriate.

A limitation of this approach is that weather from one city or station cannot perfectly represent conditions throughout an entire province. This limitation will be clearly documented and considered when interpreting model results.

Historical Scope

Although the agricultural dataset extends to 1908, the entire historical period will not automatically be used for modeling.

The climate dataset begins in 1940, and some locations have incomplete coverage during its earlier decades. Agricultural technology, crop varieties, farming practices, and markets have also changed substantially over the historical period.

During exploratory analysis, the project will examine:

- Observations available by year
- Province coverage
- Crop coverage
- Climate-data availability
- Missingness over time
- Yield trends
- Price trends

A historical cutoff will then be selected based on data coverage and relevance rather than arbitrarily choosing a year in advance.

Success Criteria

Technical Success

Regression performance will primarily be evaluated using:

Mean Absolute Error (MAE)

MAE provides an easily interpretable measure of the average prediction error in the target's original units.

Root Mean Squared Error (RMSE)

RMSE places greater emphasis on large prediction errors, which are particularly important when inaccurate yield or price predictions could substantially affect revenue estimates.

R^2

R^2 will measure how much variation in yield or price is explained by the model.

Model performance will also be compared against simple baseline predictions. A model will only be considered useful if it provides a meaningful improvement over these baselines on future, unseen data.

Business Success

Business success will be evaluated based on whether the system can:

- Generate reasonable estimates of expected yield and farm price.
- Convert predictions into expected gross revenue per hectare.
- Compare crops using a consistent financial measure.
- Identify meaningful differences in expected revenue and historical risk.
- Provide recommendations that are understandable and actionable for agricultural producers.

The best technical model will therefore not automatically be considered the best business solution if its results cannot be interpreted or translated into useful decisions.

The combined decision system will also be evaluated on its ability to estimate gross revenue per hectare on the held-out test period. Predicted yield and predicted price will be combined to calculate predicted revenue per hectare and compared with realized historical revenue per hectare. This provides an end-to-end business metric in addition to evaluating the two regression models separately. The analysis will also examine whether the system reliably distinguishes higher- and lower-revenue crop opportunities, since correct comparative ranking may be more valuable to a producer than minimizing prediction error alone.

Problem-Solving Process

1. Data Acquisition and Understanding

Both datasets will be downloaded from Kaggle and analyzed in Python using Jupyter Notebook.

Initial exploration will include:

- Dataset dimensions and data types
- Summary statistics
- Missing-value analysis
- Duplicate detection
- Crop and province frequencies
- Historical coverage
- Price and yield distributions
- Climate-data availability by year and location
- Potential outliers and implausible observations

Preliminary visualizations will include distributions, historical trend plots, crop and province comparisons, correlation analysis, and climate-versus-yield relationships.

Special attention will be given to determining the appropriate historical period for modeling.

2. Data Preparation and Feature Engineering

Data cleaning will include:

- Standardizing column names
- Handling missing values
- Checking duplicate records
- Validating numerical ranges
- Standardizing crop and geographic categories
- Converting climate dates into proper datetime values
- Investigating extreme yield, price, and climate observations

Climate data will be aggregated from daily observations into annual growing-season measures before being merged with agricultural records.

Potential engineered features include:

- Growing-season mean temperature
- Growing-season total precipitation
- Temperature variability
- Extreme-temperature counts
- Dry/wet day counts
- Previous-year crop price
- Rolling historical crop price
- Previous-year crop yield

- Rolling historical yield
- Long-term crop or provincial trends

Lagged and rolling features will be constructed using **past observations only** to prevent future information from leaking into model training.

Because both seeded-area variables represent the same underlying concept in different units, they will not automatically be included together. Their usefulness and potential redundancy will be evaluated during preprocessing.

Variables such as realized production and total farm value will also be carefully excluded from predictive stages where they would introduce information that would not have been known when the planting decision was made.

A **scikit-learn Pipeline** will be developed to ensure that transformations are applied consistently and to reduce the risk of data leakage. Numerical and categorical preprocessing will be handled within the pipeline where appropriate.

3. Modeling Strategy

Before evaluating machine learning algorithms, simple benchmark predictions will be established, such as predicting historical mean or recent rolling-average yield and price by crop. These benchmarks will provide a minimum performance standard that more complex models must outperform to justify their added complexity. Two regression tasks will be developed.

Model 1: Yield Prediction

Target: Average yield (kg/hectare)

Potential predictors include:

- Crop
- Province
- Historical year/trend
- Growing-season climate variables
- Historical/lagged yield information

The model will estimate expected agricultural productivity per hectare.

Model 2: Price Prediction

Target: Average farm price (\$/tonne)

Potential predictors include:

- Crop
- Province
- Historical trends
- Previous prices
- Rolling historical price measures
- Other information available before the prediction year

The model will estimate the expected market value of each crop.

Algorithms

At least three regression approaches will be evaluated. The initial candidates are:

1. **Linear or Ridge Regression** — provides an interpretable baseline and establishes whether relatively simple relationships can explain the targets.
2. **Random Forest Regressor** — captures nonlinear relationships and interactions while requiring fewer assumptions about the underlying relationships.
3. **Gradient Boosting Regression** — provides a more sophisticated ensemble approach capable of learning complex patterns.

The exact algorithms may be refined based on findings during EDA and initial modeling.

Validation Strategy

Because this project attempts to predict future agricultural outcomes from historical information, a conventional random train/test split could create unrealistic evaluation conditions.

Instead, the project will use a **chronological validation strategy**.

Earlier years will be used for training, later observations for validation, and the most recent period will be reserved as a final test set.

Where cross-validation is used for model comparison or hyperparameter tuning, a time-aware approach such as **TimeSeriesSplit** will be considered so that models are always trained using observations occurring before their validation observations.

This approach better represents the actual business scenario: using information available today to make predictions about future agricultural outcomes.

Hyperparameter Tuning

Promising models will be tuned using approaches such as [RandomizedSearchCV](#) or [GridSearchCV](#), combined with the time-aware validation strategy.

Hyperparameter tuning will focus on meaningful improvements rather than exhaustive searching, helping keep the project computationally realistic within the three-week timeframe.

4. Revenue and Risk Analysis

After the yield and price models are finalized, their predictions will be combined.

Predicted yield will first be converted from kilograms to metric tonnes per hectare. Expected gross revenue per hectare will then be calculated as:

Expected Revenue per Hectare = Predicted Yield (tonnes/hectare) × Predicted Price (\$/tonne)

This creates a common financial measure for comparing crops.

Historical variability in yield, price, and/or revenue will then be examined to represent **risk**.

For example, a crop may have the highest expected revenue but also exhibit substantially greater historical volatility. Another crop may provide slightly lower expected revenue while offering more stable historical outcomes.

The final recommendations will therefore consider both:

Expected Return + Historical Risk

rather than simply recommending whichever crop has the highest predicted revenue.

5. Results Interpretation and Communication

Model results will be translated into business language rather than presented solely through statistical metrics.

Potential final visualizations include:

- Actual vs. predicted yield
- Actual vs. predicted price
- Model-error comparisons
- Feature-importance charts
- Historical yield and price trends
- Climate relationships with yield
- Predicted revenue per hectare by crop
- Revenue versus risk comparisons

Where appropriate, permutation importance or other model-interpretation techniques will be used to identify influential predictors.

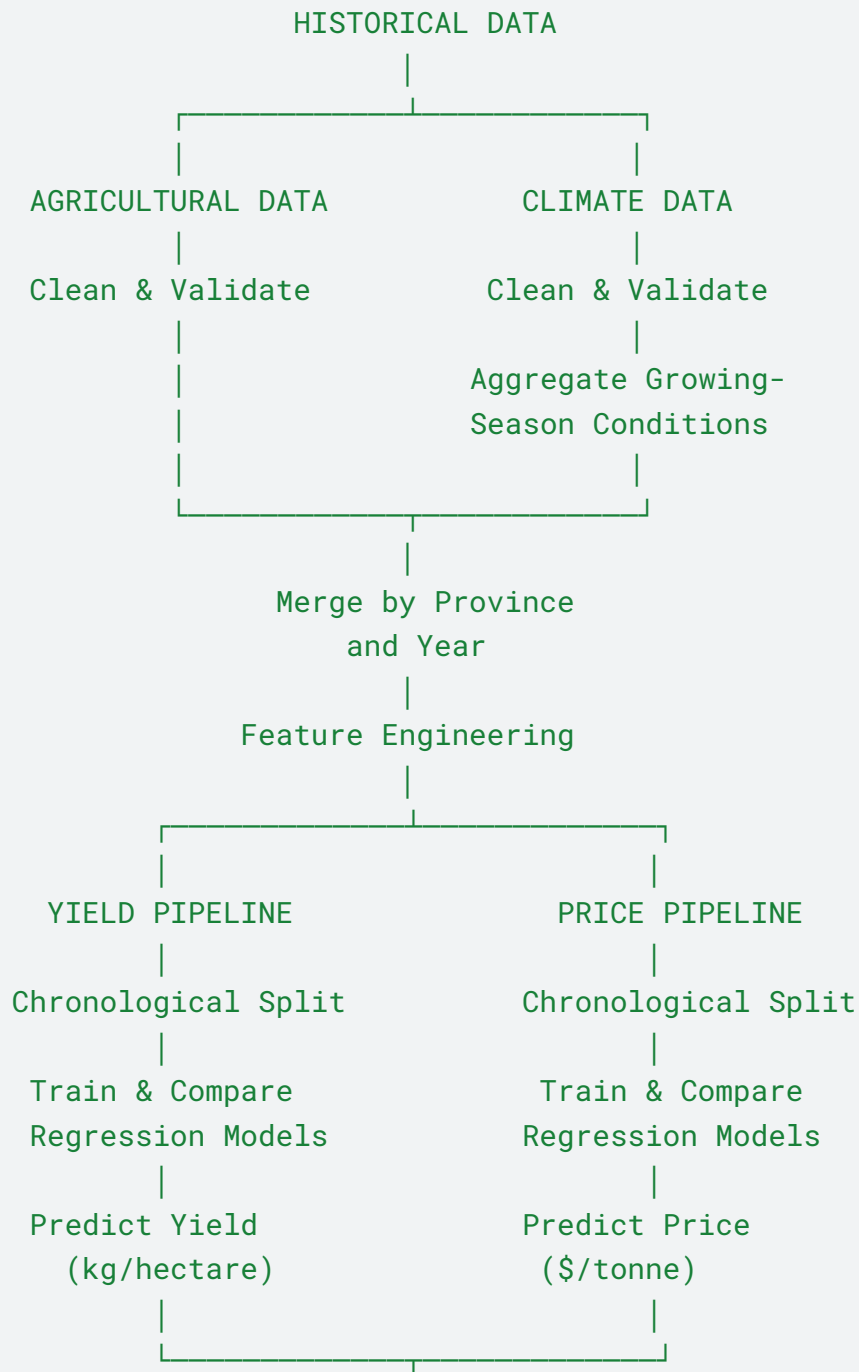
Technical concepts will be explained in terms of their business meaning. For example, RMSE will not only be reported numerically but interpreted as the approximate magnitude of prediction errors and their potential implications for revenue planning.

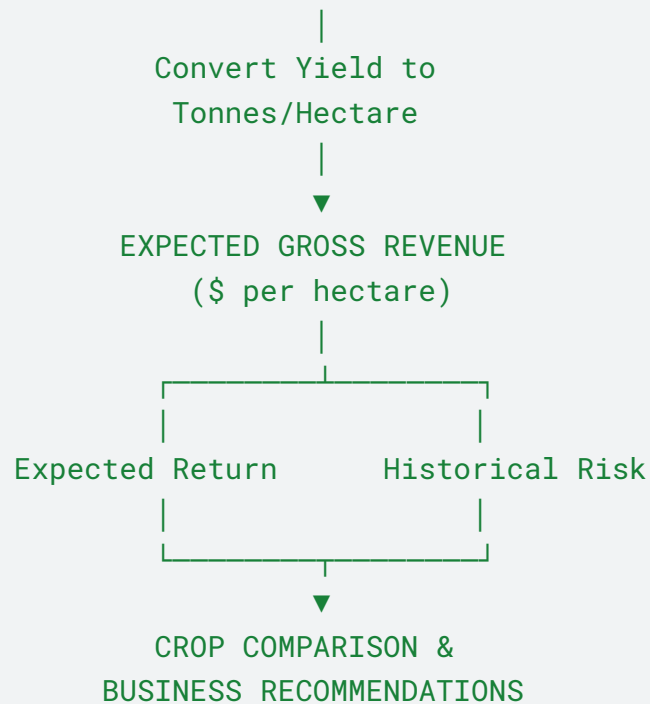
The final video presentation will focus on the business problem, major analytical findings, model performance, recommendations, limitations, and potential next steps rather than presenting code in detail.

Conceptual Framework

The proposed solution follows two predictive paths that converge into a single agricultural decision-support system:

None





Stage Dependencies

The climate dataset must first be cleaned and aggregated before it can be joined to the agricultural dataset.

EDA must be completed before selecting the historical cutoff and final features.

Feature engineering must occur before pipeline construction and model development.

Models must be validated chronologically before final predictions can be considered reliable.

Both the yield and price models must be completed before expected revenue can be calculated.

Risk analysis and model interpretation will then be combined with predicted revenue to produce final business recommendations.

Three-Week Project Timeline

The three-week timeframe provides sufficient time to complete the project without expanding the scope unnecessarily. Work will progress sequentially, while allowing earlier stages to be revisited if modeling reveals data-quality or feature issues.

Dataset Finalization and Problem Formulation — Days 1–2

- Finalize agricultural and climate datasets
- Refine business problem and stakeholder needs
- Confirm modeling targets
- Establish project repository
- Organize Jupyter Notebook structure
- Verify historical and geographic overlap between datasets

Deliverable: Finalized project scope and reproducible project environment.

Exploratory Data Analysis — Days 3–5

- Comprehensive data profiling
- Missing-value assessment
- Historical coverage analysis
- Crop and province comparisons
- Price and yield trend analysis
- Climate exploration
- Statistical analysis of important relationships
- Determine justified historical cutoff
- Document findings and limitations

Deliverable: Documented EDA and finalized modeling population.

Data Preprocessing — Days 6–8

- Clean agricultural data
- Clean and reshape climate data
- Engineer growing-season climate variables
- Map weather locations to provinces
- Merge datasets
- Engineer lagged and rolling historical variables
- Identify leakage-prone variables

- Develop scikit-learn preprocessing pipeline
- Establish chronological train/validation/test periods

Deliverable: Modeling-ready dataset and preprocessing pipeline.

Model Development — Days 9–12

- Develop baseline models
- Train initial regression algorithms
- Compare model performance
- Implement time-aware cross-validation
- Conduct targeted hyperparameter tuning
- Document modeling decisions

Deliverable: Candidate yield and price models with validated performance estimates.

Model Evaluation and Refinement — Days 13–15

- Select final yield model
- Select final price model
- Evaluate both models on held-out test data
- Calculate expected revenue per hectare
- Conduct risk/volatility analysis
- Analyze feature importance
- Investigate important model errors
- Translate results into preliminary business recommendations

Deliverable: Final predictive system and business analysis.

Documentation and Reporting — Days 16–18

- Clean and comment notebook code
- Verify reproducibility
- Finalize visualizations
- Document assumptions and limitations
- Develop presentation narrative
- Prepare executive-focused presentation materials

Deliverable: Completed notebook and presentation materials.

Final Review and Submission — Days 19–21

- Conduct quality-assurance review
- Verify Git repository
- Re-run notebook from beginning to end
- Check final metrics and visualizations
- Record video presentation
- Review video and submission materials
- Submit final project

Deliverable: Complete capstone submission.

To keep the three-week project scope realistic, the project will focus on gross revenue rather than full farm profitability and will not attempt to build a complete commodity-market forecasting system. Additional external datasets will only be incorporated if they provide a clear improvement to the core yield or price models without jeopardizing the project timeline.

Potential Challenges and Additional Research

Geographic Representation of Climate Data

The climate dataset contains observations from individual cities rather than province-wide measurements. A weather station may not fully represent growing conditions across an entire province.

The project will therefore document station-to-province mappings and treat this as a limitation. Additional research may be required to determine whether particular stations provide reasonable proxies for major agricultural areas.

Historical Data Coverage

The agricultural and climate datasets cover different historical periods, and climate availability varies by location. EDA will be required to identify a period with sufficient overlap and reliable coverage.

Changing Agricultural Conditions

Farming technology, crop genetics, production practices, and markets have changed considerably over the historical period. Older observations may therefore be less representative of current agriculture. The historical cutoff will be selected based on both data quality and business relevance.

Price Changes and Inflation

Long-term nominal price comparisons may be distorted by inflation. Additional research may be conducted to determine whether inflation adjustment is necessary for historical financial analysis. For prediction, emphasis may instead be placed on lagged prices, relative changes, and recent historical information.

Data Leakage

Several variables are closely related mathematically. Production depends on yield and planted area, while total farm value is closely related to production and realized price.

Including these variables indiscriminately could produce unrealistically strong model performance. Feature selection will therefore be based on what information would realistically be available when a planting decision is made.

Price Forecasting Difficulty

Agricultural prices are influenced by factors beyond the available dataset, including international commodity markets, supply and demand, trade conditions, and macroeconomic factors. The price model may therefore have greater uncertainty than the yield model.

This limitation will be evaluated through model performance rather than hidden, and the project's conclusions will reflect the level of predictive accuracy actually achieved.

Revenue Versus Profit

The project estimates gross revenue rather than net profit because crop-specific production costs are unavailable. Recommendations will therefore be presented as revenue and risk comparisons rather than claims about which crop will necessarily produce the greatest profit.

Expected Final Outcome

The final product will be an end-to-end machine learning solution that combines agricultural and climate data to predict crop yield and farm prices and translates those predictions into expected gross revenue per hectare.

Rather than providing producers with a single prediction, the project will create a framework for comparing crop opportunities based on both **expected economic return and historical uncertainty**.

The final analysis will demonstrate the complete data science lifecycle, including business understanding, data acquisition, exploratory analysis, preprocessing, feature engineering, machine learning pipelines, regression modeling, chronological validation, model evaluation, interpretation, and communication of actionable business insights.